Software Engineering Project Report

Almirah E-Commerce Platform
National University of Computer and Emerging Sciences
Course: SE-3009 (Software Engineering)
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1. Introduction

The Almirah E-Commerce Platform is a web-based application designed to modernize the online shopping experience for Almirah, a renowned Pakistani fashion brand. It aims to provide a seamless, user-friendly interface for browsing, purchasing, and managing orders, enhancing customer engagement and satisfaction.

2. Project Rationale

The current Almirah website suffers from usability issues, including poor navigation, inconsistent design, limited search functionality, and lack of accessibility features. These challenges reduce user satisfaction and conversion rates. Our project addresses these by delivering a redesigned website with intuitive navigation, enhanced search capabilities, and accessibility compliance, ensuring an inclusive and efficient shopping experience. By adopting agile methodologies (Scrum), we ensure iterative development, continuous feedback, and alignment with stakeholder needs.

3. Project Architecture

Architecture Style

The Almirah E-Commerce Platform employs a **three-tier layered architecture** consisting of:

- **Presentation Layer**: Built using HTML, CSS, JavaScript, and React for a responsive and dynamic user interface.
- **Business Logic Layer**: Implemented with Node.js to handle server-side logic, API integrations, and user interactions.
- **Data Layer**: Utilizes MongoDB for efficient storage and retrieval of product, user, and order data.

Justification

The three-tier architecture was chosen for the following reasons:

- **Modularity**: Each layer operates independently, allowing for easier maintenance and updates.
- **Scalability**: The separation of concerns enables horizontal scaling (e.g., adding more servers for the business logic layer).
- **Maintainability**: Developers can work on different layers simultaneously, improving team efficiency.
- **Security**: Isolating the data layer enhances protection against unauthorized access.
- **Flexibility**: The architecture supports future integrations, such as mobile apps or additional payment gateways.

This structure aligns with industry standards for e-commerce platforms, balancing performance and development efficiency.

4. Product Backlog

The product backlog outlines all user stories prioritized for development, with estimates and acceptance criteria.

ID	User Story	Estimate	Priority	Acceptance Criteria
1	As a new user, I want to sign up using my email and password so that I can create an account.	6 hours	1	- User can register with valid email and password. - Account is stored in the database. - User receives a confirmation email.
2	As a returning user, I want to log in using my credentials so that I can access my saved information.	4 hours	1	- User can log in with correct credentials. - Session is created securely. - Invalid credentials display an error.
3	As a user, I want to reset my password so that I can regain access if I forget it.	3 hours	2	- User can request a password reset via email. - A secure reset link is sent. - Password is updated successfully.
4	As a user, I want to browse products by categories so that I	8 hours	1	- Products are displayed by categories (e.g., Men’s

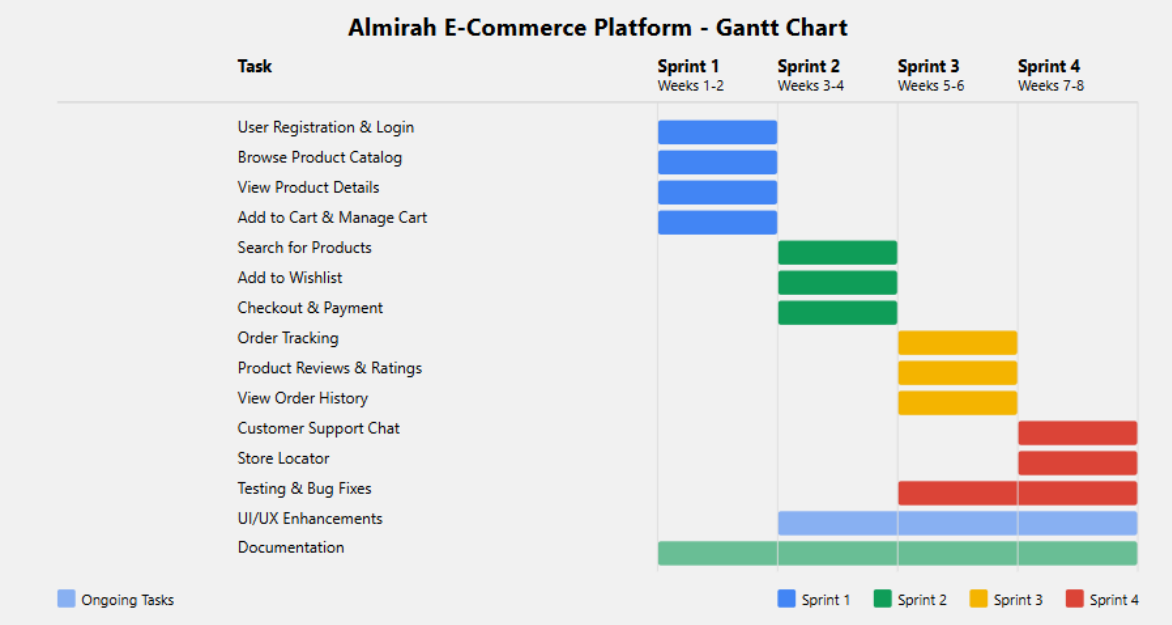
ID	User Story	Estimate	Priority	Acceptance Criteria
	5 can easily find what I am looking for.			Wear, Women's Wear). - Categories are filterable and navigable.
5	As a user, I want to see a list of trending or featured products so that I can explore popular items.	4 hours	3	- Trending products are displayed on the homepage. - List is dynamically updated based on sales data.
6	As a user, I want to search for products using a keyword so that I can quickly find what I need.	6 hours	1	- Search returns relevant results based on keywords. - Results load within 2 seconds.
7	As a user, I want to filter search results by price, category, and availability so that I can refine my choices.	5 hours	2	- Filters for price, category, and availability are functional. - Filtered results update dynamically.
8	As a user, I want to see detailed product descriptions and high-quality images so that I can make an informed purchase decision.	7 hours	1	- Product page shows images, descriptions, sizes, and prices. - Images are zoomable.
9	As a user, I want to zoom in on product images so that I can see fabric details.	3 hours	3	- Zoom functionality is available on product images. - Images remain clear when zoomed.
10	As a user, I want to add products to my wishlist so that I can save them for future reference.	4 hours	4	- User can add/remove products to/from wishlist. - Wishlist is accessible when logged in.
11	As a user, I want to view my wishlist anytime so that I can decide whether to purchase the items later.	3 hours	4	- Wishlist displays all saved items. - User can navigate to product pages from wishlist.

ID	User Story	Estimate	Priority	Acceptance Criteria
12	As a user, I want to add selected items to my shopping cart so that I can purchase multiple items at once.	5 hours	1	- Products can be added to cart. - Cart updates dynamically.
13	As a user, I want to update item quantities in my cart so that I can adjust my order before checkout.	4 hours	1	- User can increase/decrease quantities. - Cart reflects updated prices.
14	As a user, I want to remove items from my cart so that I can modify my purchase before payment.	3 hours	1	- User can remove items from cart. - Cart updates total price accordingly.
15	As a user, I want to enter my shipping details so that I can receive my order at the correct address.	5 hours	2	- User can input valid shipping details. - Details are saved for checkout.
16	As a user, I want to choose my preferred payment method so that I can complete my purchase securely.	6 hours	2	- Payment options (e.g., credit card, cash on delivery) are available. - Transaction is secure.
17	As a user, I want to receive an order confirmation email so that I have proof of my purchase.	3 hours	2	- Order confirmation email is sent post-purchase. - Email includes order details.
18	As a user, I want to track my order status so that I know when it will be delivered.	5 hours	3	- Order tracking page shows real-time status. - Updates are accurate.
19	As a user, I want to receive SMS/email updates about my order so that I stay informed about the shipping progress.	4 hours	3	- Notifications are sent for order updates. - User can opt out of notifications.

ID	User Story	Estimate	Priority	Acceptance Criteria
20	As a user, I want to rate products I have purchased so that I can share my feedback with others.	4 hours	4	- Users can submit ratings post-purchase. - Ratings are displayed on product pages.
21	As a user, I want to read customer reviews so that I can make better purchasing decisions.	3 hours	4	- Reviews are visible on product pages. - Reviews are authentic and moderated.
22	As a user, I want to chat with customer support so that I can resolve any issues with my order.	8 hours	5	- Live chat is available during business hours. - Responses are timely.
23	As a user, I want to receive automated responses from a chatbot so that I can get quick answers to common questions.	7 hours	5	- Chatbot handles common queries. - Escalates complex issues to human support.
24	As a user, I want to find the nearest Almirah store so that I can visit in person.	4 hours	5	- Store locator shows nearby stores with addresses. - Map integration is functional.
25	As a user, I want to see store contact details so that I can call for inquiries before visiting.	3 hours	5	- Store contact details are accurate. - Details are accessible from store locator.
26	As a registered user, I want to see my past orders so that I can track my purchase history.	4 hours	4	- Order history is accessible when logged in. - Details include order date and status.
27	As a user, I want to reorder previously purchased items so that I can buy my favorite products again easily.	3 hours	4	- Reorder option is available in order history. - Items are added to cart seamlessly.

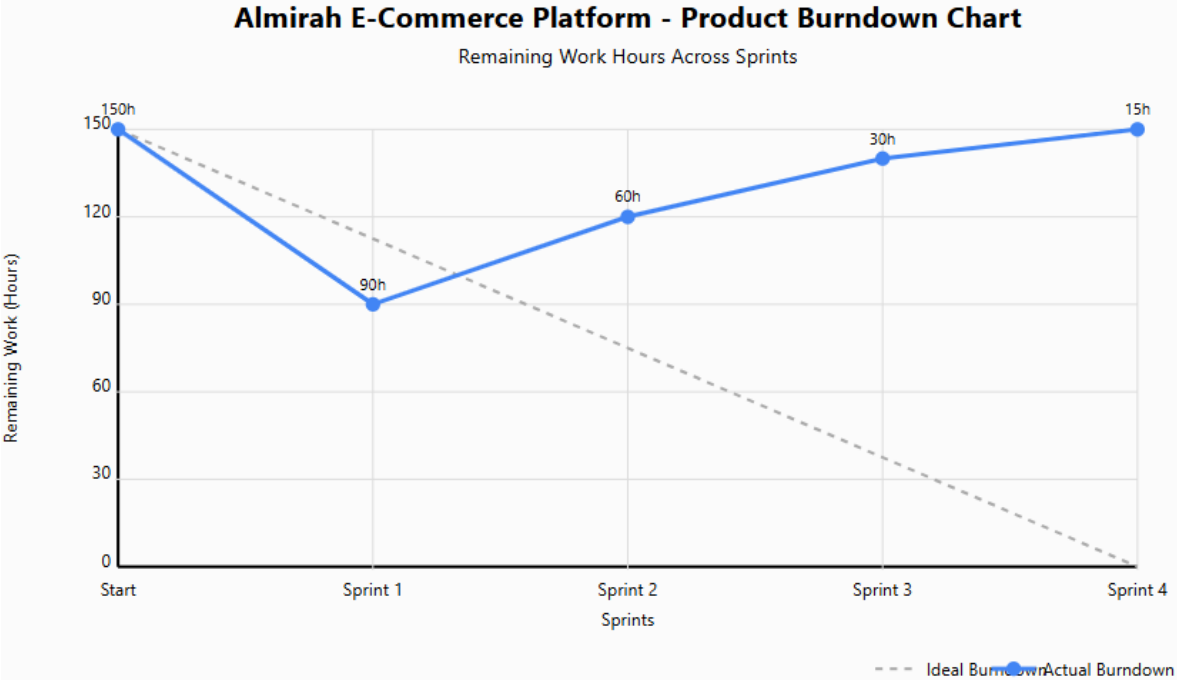
5. Project Planner and Gantt Chart

The project is divided into multiple sprints, each lasting 1-2 weeks.



6. Product Burndown Chart

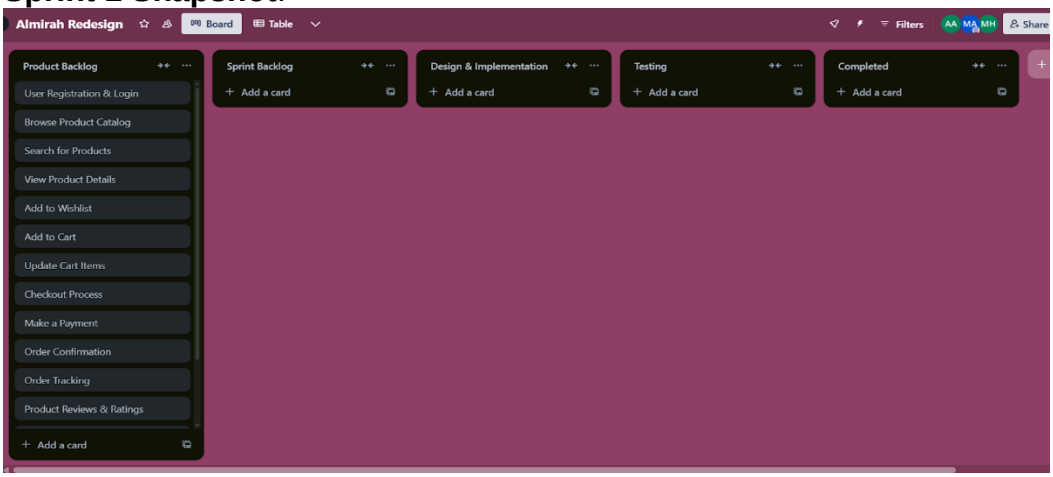
The burndown chart tracks the remaining work across sprints.



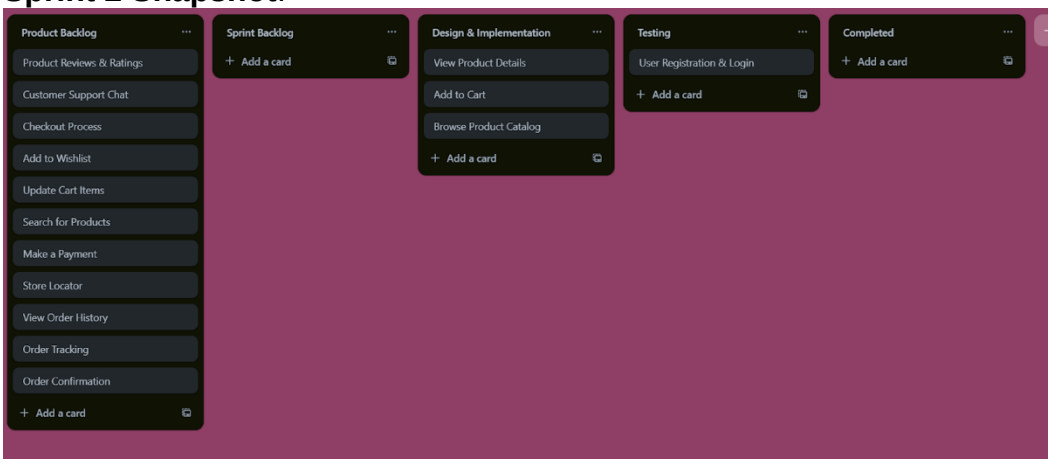
7. Trello Snapshots

Below are snapshots of the Trello board at the end of each sprint and at project completion.

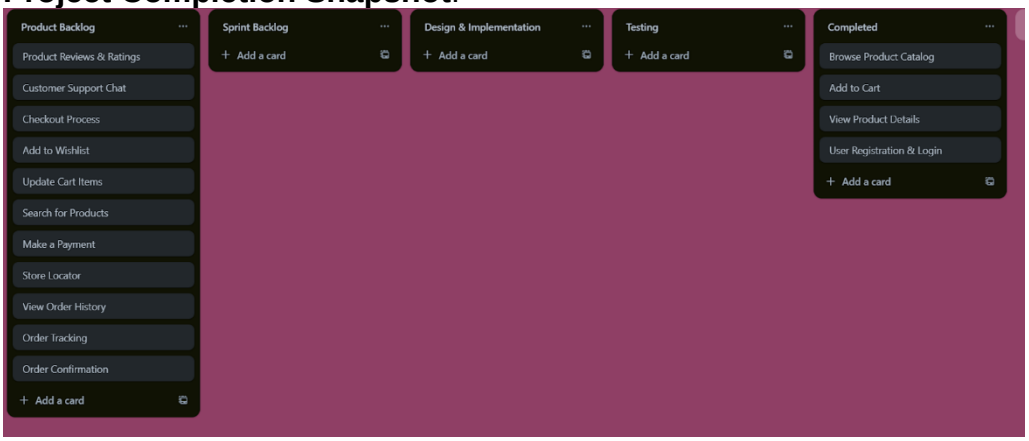
- **Sprint 1 Snapshot**:



- **Sprint 2 Snapshot**:



- **Project Completion Snapshot**:



8. GUIs and Testing

The Almirah website includes the following GUI modules, each tested with boundary value analysis (BVA) and equivalence class partitioning (ECP).

GUI Modules

1. **User Registration & Login**
    - Fields: Email, Password, Name, Phone Number
    - Features: Sign-up, login, password reset
2. **Browse Product Catalog**
    - Fields: Category Dropdown, Filter Buttons (Price, Brand)
    - Features: Category browsing, trending products
3. **Search for Products**
    - Fields: Search Bar, Filter Options (Price, Category, Availability)
    - Features: Keyword search, result filtering
4. **View Product Details**
    - Fields: Product Image, Description, Size Selector, Price
    - Features: Zoomable images, product reviews
5. **Add to Cart & Manage Cart**
    - Fields: Quantity Input, Remove Button, Total Price
    - Features: Add/remove items, update quantities

Testing

User Registration (User Story 1 & 2)

Equivalent Class Partitions

Input	Valid Equivalence Classes	Invalid Equivalence Classes
Name	Alphabetic, ≥3 chars	Non-alphabetic, <3 chars
Email	Contains @ and .com	No @ or .com
Password	6-10 chars, includes uppercase, lowercase, special char	<6 or >10 chars, missing required chars
Phone Number	7-15 digits, numeric only	<7 or >15 digits, non-numeric

Boundary Value Analysis

Input	Boundary Values
Name	3 chars
Password	6 chars, 10 chars
Phone Number	7 digits, 15 digits

Search for Products (User Story 6 & 7)

Equivalent Class Partitions

Input	Valid Equivalence Classes	Invalid Equivalence Classes
Search Keyword	Alphanumeric, ≥1 char	Empty input
Price Filter	Numeric range (e.g., 1000-50000)	Non-numeric, negative values
Category Filter	Valid category selected	No category selected

Boundary Value Analysis

Input	Boundary Values
Search Keyword	1 char
Price Filter	1000, 50000

Add to Cart (User Story 12-14)

Equivalent Class Partitions

Input	Valid Equivalence Classes	Invalid Equivalence Classes
Quantity	Integer, 1-10	Non-integer, <1, >10
Remove Button	Clicked	Not clicked

Boundary Value Analysis

Input	Boundary Values
Quantity	1, 10

9. Product Review

Improvements

- **UI Enhancement**: Transitioning to a more modular framework like React with Tailwind CSS could improve UI reusability and responsiveness.
- **Performance**: Caching search results and optimizing database queries could reduce response times below 2 seconds.
- **Mobile App**: Developing a companion mobile app would expand accessibility.

Limitations

- **Scalability**: The current architecture may face performance bottlenecks under very high traffic (e.g., >1000 concurrent users).
- **Accessibility**: While basic accessibility features are implemented, advanced features like screen reader optimizations are incomplete.
- **Payment Options**: Limited to credit card and cash on delivery; integrating mobile wallets (e.g., JazzCash) would enhance user convenience.